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Timothy R. Langlois

education

Ph.D., Computer Science, Cornell University, 2011–2016.
Advised by Doug James

B.S., Computer Systems Engineering, University of Massachusetts Amherst, 2009.
Summa Cum Laude
Double major in Computer Science

research interests

Computer Graphics
Computational Physics
Acoustics
Scientific Computing

publications

Conference and Journal Articles

Yixin Chen, David Levin, and Timothy R. Langlois "Fluid Control with Localized Spacetime Windows." *In Submission*. 2026.

Jialin Huang, Prem Seetharaman, Timothy R. Langlois, Li-Yi Wei, Rubaiat Habib Kazi, and Yotam Gingold "MoSound: An Interactive Tool for Generative Sound Design in Motion Graphics." *CHI 2026*. 2026.

Zeqi Gu, Difan Liu, Timothy Langlois, Matthew Fisher, and Abe Davis "How to Train Your Dragon: Automatic Diffusion-Based Rigging for Characters with Diverse Topologies." *Eurographics 2025*. 2025.

Vivian Liu, Rubaiat Habib Kazi, Li-Yi Wei, Matt Fisher, Timothy Langlois, Seth Walker, and Lydia Chilton "LogoMotion: Visually-Grounded Code Generation for Content-Aware Animation." *CHI 2025*. 2025.

Kangrui Xue, Jui-Hsien Wang, Timothy R. Langlois, and Doug L. James "WaveBlender: Practical Sound-Source Animation in Blended Domains." *ACM Transactions on Graphics (SIGGRAPH Asia 2024)*. 2024.

Yixin Chen, David Levin, and Timothy R. Langlois "Fluid Control with Laplacian Eigenfunctions." *ACM Transactions on Graphics (SIGGRAPH 2024)*. 2024.

Kangrui Xue, Ryan M. Aronson, Jui-Hsien Wang, Timothy R. Langlois, and Doug L. James "Improved Water Sound Synthesis using Coupled Bubbles." *ACM Transactions on Graphics (SIGGRAPH 2023)*. 2023.

Qiaodong Cui, Timothy R. Langlois, Pradeep Sen, and Theodore Kim "Spiral-spectral fluid simulation." *ACM Transactions on Graphics (SIGGRAPH Asia 2021)*. 2021.

Danyong Zhao, Yijing Li, Siddhartha Chaudhuri, Timothy R. Langlois and Jernej Barbic "ERGOBOSS: Ergonomic Optimization of Body-Supporting Surfaces." *IEEE Transactions on Visualization and Computer Graphics*. 2021.

Zachary Ferguson, Minchen Li, Teseo Schneider, Francisca Gil-Ureta, Timothy R. Langlois, Chenfanfu Jiang, Denis Zorin, Danny M. Kaufman, and Daniele Panozzo "Intersection-free Rigid Body Dynamics." *ACM Transactions on Graphics (SIGGRAPH 2021)*. August, 2021.

Minchen Li, Zachary Ferguson, Teseo Schneider, Timothy R. Langlois, Denis Zorin, Daniele Panozzo, Chenfanfu Jiang, and Danny M. Kaufman "Incremental Potential Contact: Intersection- and Inversion-free, Large-Deformation Dynamics." *ACM Transactions on Graphics (SIGGRAPH 2020)*. August, 2020.

Qiaodong Cui, Timothy R. Langlois, Pradeep Sen, and Theodore Kim "Fast and Robust Stochastic Structural Optimization." *Computer Graphics Forum (Eurographics)*. 2020.

Zhenyu Tang, Nicholas J. Bryan, Dingzeyu Li, Timothy R. Langlois, and Dinesh Manocha "Scene-Aware Audio Rendering via Deep Acoustic Analysis." *IEEE VR 2020 Journal Track (TVCG)*. March, 2020.

Minchen Li, Ming Gao, Timothy R. Langlois, Chenfanfu Jiang, Danny Kaufman "Decomposed Optimization Time Ingetrator for Large-Step Elastodynamics." *ACM Transactions on Graphics (SIGGRAPH 2019)*. August, 2019.

Rahul Arora, Alec Jacobson, Timothy R. Langlois, Yijiang Huang, Caitlin Mueller, Wojciech Matusik, Ariel Shamir, Karan Singh, David I.W. Levin "Volumetric Michell Trusses for Parametric Design & Fabrication." *Proceedings of the 3rd ACM Symposium on Computational Fabrication (SCF)*. June, 2019.

Pedro Morgado, Nuno Vasconcelos, Timothy R. Langlois, Oliver Wang. "Self-Supervised Generation of Spatial Audio for 360° Video." *Neural Information Processing Systems (NIPS)*. December, 2018.

Jui-Hsien Wang, Ante Qu, Timothy R. Langlois, Doug James. "Toward Wave-based Sound Synthesis for Computer Animation." *ACM Transactions on Graphics (SIGGRAPH 2018)*. 37(4), August, 2018.

Dingzeyu Li, Timothy R. Langlois, Changxi Zheng. "Scene-Aware Audio for 360° Videos." *ACM Transactions on Graphics (SIGGRAPH 2018)*. 37(4), August, 2018.

Timothy R. Langlois, Ariel Shamir, Daniel Dror, Wojciech Matusik, David I.W. Levin. "Stochastic Structural Analysis for Context-Aware Design and Fabrication." *ACM Transactions on Graphics (SIGGRAPH Asia 2016)*. 35(6), December, 2016.

Timothy R. Langlois, Changxi Zheng, and Doug L. James. "Toward Animating Water with Complex Acoustic Bubbles." *ACM Transactions on Graphics (SIGGRAPH 2016)*. 35(4), July, 2016.

Timothy R. Langlois, Steven S. An, Kelvin K. Jin, and Doug L. James. "Eigenmode Compression for Modal Sound Models." *ACM Transactions on Graphics (SIGGRAPH 2014)*. 33(4), August, 2014.

Timothy R. Langlois and Doug L. James. "Inverse-Foley Animation: Synchronizing rigid-body motions to sound." *ACM Transactions on Graphics (SIGGRAPH 2014)*. 33(4), August, 2014.

Timothy R. Langlois, Ramgopal R. Mettu, and Richard W. Vachet. "Protein Identification Using Receptor Arrays and Mass Spectrometry." *Advances in Computational Biology*, Springer Advances in Experimental Medicine and Biology Series. 680, 343-351, 2010.

Patents

Rubaiat Habib Kazi, Timothy Langlois, Li-Yi Wei, John Simpson, James Corbett, Christopher Nuuja, Brooke Hopper. "Dynamic path animation of animation layers and digital design objects." U.S. Patent No. 12,051,143. 30 July 2024.

Rubaiat Habib Kazi, Timothy Langlois, Li-Yi Wei, John Simpson, James Corbett, Christopher Nuuja, Brooke Hopper. "Providing and utilizing a one-dimensional layer motion element to generate and manage digital animations." U.S. Patent No. 11,967,011. 23 April 2024.

Zhenyu Tang, Timothy Langlois, Nicholas Bryan, and Dingzeyu Li. "Generating scene-aware audio using a neural network-based acoustic analysis." U.S. Patent No. 11,812,254. 07 Nov 2023.

Zhenyu Tang, Timothy Langlois, Nicholas Bryan, and Dingzeyu Li. "Rendering scene-aware audio using neural network-based acoustic analysis." U.S. Patent No. 11,190,898. 30 Nov 2021.

Oliver Wang, Pedro Morgado, and Timothy Langlois. "Generating spatial audio using a predictive model." U.S. Patent No. 10,701,303. 30 June 2020.

David Levin, Daniel Dror, Timothy Langlois, Wojciech Matusik, and Ariel Shamir. "Stochastic structural analysis for context-aware design and fabrication." U.S. Patent App No. 15/837,610. 14 June 2018.

Srinivas Ravela, William J. Dupree, Timothy R. Langlois, Marilyn M. Wolfson, and Christopher M. Yang. "Method and apparatus for generating a forecast weather image." U.S. Patent No. 8,625,840. 7 Jan. 2014.

experience

Senior Research Scientist 2: Adobe Research, 2022-Present.

Senior Research Scientist: Adobe Research, 2018-2022.

Research Scientist: Adobe Research, 2016-2018.

Research Intern: Disney Research Boston, 2015 (summer).

Software Engineer: MIT Lincoln Laboratory, 2009–2011.
Developed distributed, real-time weather prediction algorithms.

Software Engineering Intern: Raytheon, 2008 (summer).

Software Engineering Intern: DEKA Research and Development, 2006–2008 (summers/winters).
Developed software for embedded systems on various medical devices.

teaching

Teaching Assistant: Physically Based Animation (CS5643), Cornell University, 2015.

Teaching Assistant: Introduction to Computer Graphics (CS4620), Cornell University, 2013.

service

Paper Committes:

- SIGGRAPH
- SIGGRAPH Asia
- SCA

Reviewer:

- ACM SIGGRAPH
- ACM SIGGRAPH Asia
- ACM TON
- CGF
- ACM TOG
- TVCG
- ECCV
- Pacific Graphics
- IEEE VR
- SCA

Invited Talks:

- Graphyz 2026 Keynote

Volunteer: Adobe Employee Community Fund Grant Committee, 2018, 2021.
Reviewed applications from nonprofit organizations for Adobe grants.

Organizer: Cornell Computer Science Student Brown Bag Seminar, 2013-2015.
Organized weekly presentations on current research by graduate students.

Volunteer: Expand Your Horizons, 2012.
Co-organized an educational workshop for middle-school students.

grants & awards

Recipient: National Science Foundation, Graduate Research Fellowship, 2012–2016.